

Fundamentals of Business & Financial Analysis

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Fundamentals of Business & Financial Analysis

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I. What Drives a Business?

STARTING A BUSINESS

It all starts with an idea...

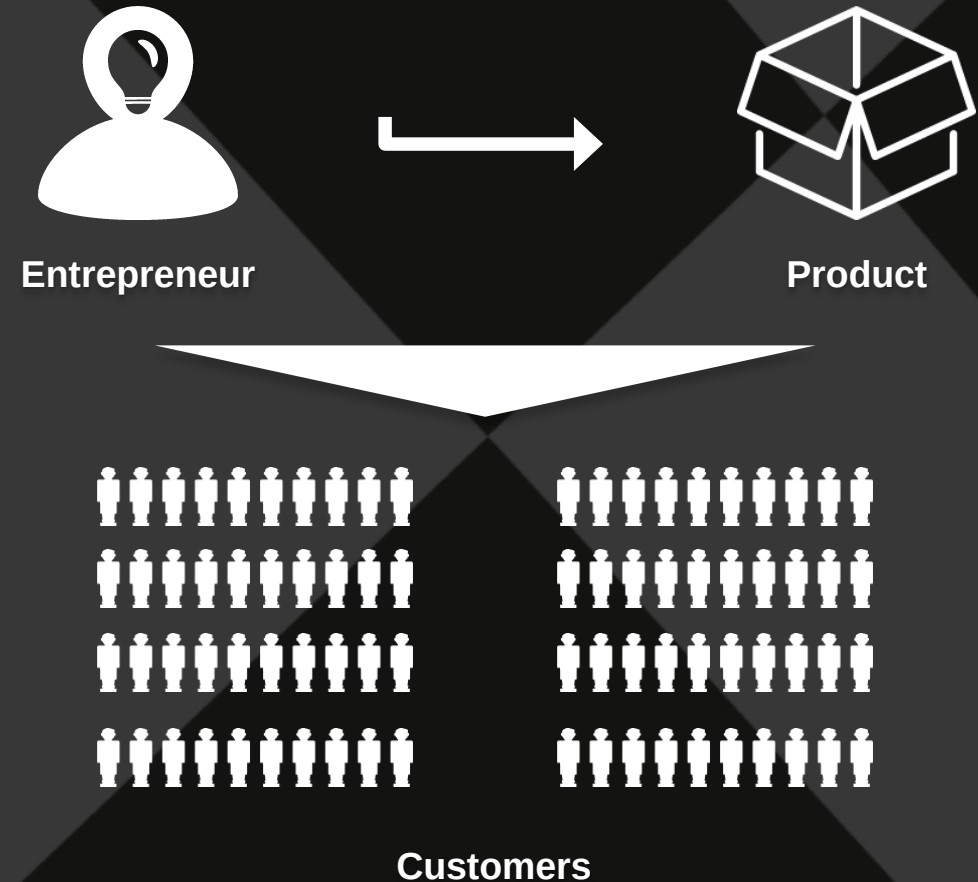
Entrepreneurs start a business, because they want to offer a new product to their customers

They also believe a large enough number of customers will be interested in their product, which would allow them to make sales and stay in business

Building a successful business is not easy, as there are a number of competitor companies that are trying to achieve the same goal: selling their goods to customers

In order to be successful, a firm needs to find ways to create value for its customers

This is the crucial question every entrepreneur needs to think about: “How is my company creating value for its customers?”



STAYING IN BUSINESS

So what are the goals of a company after it is founded?

The primary goal for a company should always be to create great products that satisfy customers' needs. Business is about making clients happy

Then, the firm needs to sell its products to a sufficient number of clients and earn revenues

These revenues can be reinvested in the company's business, and it can start expanding

But, doing business is not just about making money. It is also about empowering the firm's employees and contributing to society, in general



Create great products



Sell products to customers



Make money and reinvest part of it in the business



Satisfy stakeholders: employees, ownership, society

II. Making Money

WHEN IS A BUSINESS PROFITABLE?

Being profitable – the goal of every business

Profit is the money a firm makes after accounting for all of its expenses. This is what we see in the equation on the right:

Revenues (money coming in after selling goods to customers), minus Costs (money going out for salaries, raw materials, utilities, taxes, etc.) equals a Profit or Loss

This equation calculates whether a company is making or losing money

REVENUES – COSTS = PROFIT/LOSS



ARE THERE UNPROFITABLE COMPANIES?

Yes, there are.

It is difficult to build a profitable business.

On the right side you can see the logos of some large and very well-known companies. However, these companies have struggled to produce profits in the past several years

It is important to distinguish between young and mature companies that are unprofitable

Most start-up businesses (young companies) are loss-making, because they need some time before expanding their business and covering fixed costs. On the other hand it is very alarming if a mature business starts losing money



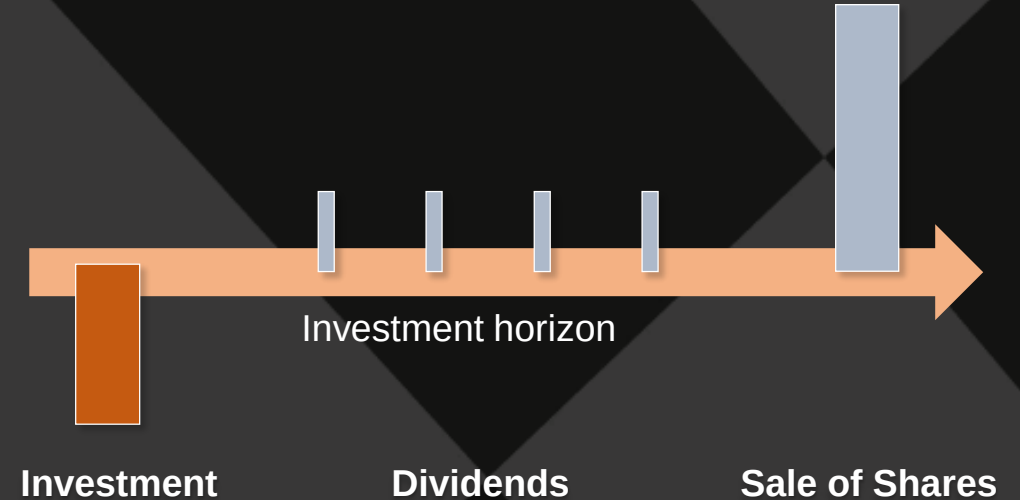
WHY INVESTORS INVEST IN COMPANIES?

Investors give their money and expect much more in return

All investors are interested in obtaining a great return on their investment. They expect a certain premium, due to the fact they put their money in a company (which is risky) compared to investing it in treasury bonds (which is virtually risk-free)

When investors put their money in a company, they experience a cash outflow. Throughout the life of the company, they will receive dividends; hence, they will have a cash inflow. At some point most investors sell their shares in a particular company, and this represents another cash inflow

This is the reasons investors value companies, based on the amount of cash they expect to receive from these companies. At the same time, the expected amount of cash they will receive is directly linked to a company's profitability and growth



An investor's primary goal?
Return on Investment

WHAT DRIVES COMPANY VALUE?

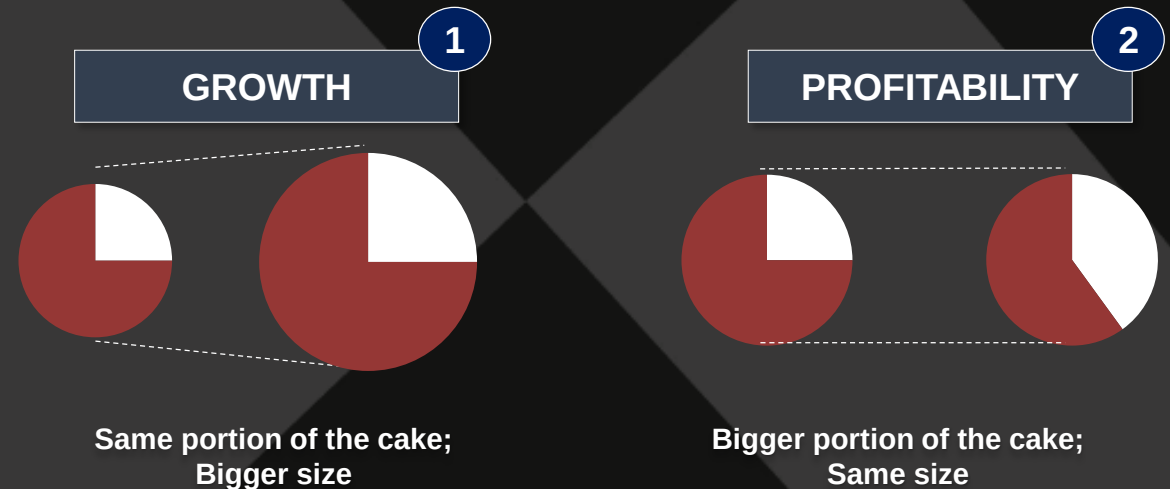
A business becomes more valuable when it is:
i) growing and/or ii) becoming more profitable

Given that investors are interested in a company's profits we have to be aware of what a company can do in order to grow its profits

It either has to grow its business, thus earning more money by selling more products to more customers, or it has to make its business more profitable

Measures like Gross Margin and EBITDA% look into a company's profitability – its ability to generate profits after it sells products to clients. Let's provide an easy example. Think of two companies. Both sell only one product - a cell phone. Company A sells a phone for \$200 and sustains costs of \$50 in order to produce one phone. Company B sells its product at the same price (\$200), but sustains \$80 of costs in order to produce it. It is obvious that Company A makes more money from each unit that it sells

Two functions drive a company's value:



WHAT DRIVES COMPANY VALUE?

Let's take a look at a practical example!



On the right we see Nike's 3 year Profit & Loss for the years between 2016 and 2018. The company was able to increase its sales by an average of 6% per year. This represents the growth component we saw in the previous slide. This was mainly due to increasing Footwear and Apparel sales

At the same time its EBIT margin, showing operating profitability decreased from 13.9% in 2016 to 12.2% in 2018.

Two functions drive a company's value:

Results of Operations

(Dollars in millions, except per share data)	Fiscal 2018	Fiscal 2017	% Change	Fiscal 2016	% Change
Revenues	\$ 36,397	\$ 34,350	6%	\$ 32,376	6%
Cost of sales	20,441	19,038	7%	17,405	9%
Gross profit	15,956	15,312	4%	14,971	2%
Gross margin	43.8%	44.6%		46.2%	
Demand creation expense	3,577	3,341	7%	3,278	2%
Operating overhead expense	7,934	7,222	10%	7,191	0%
Total selling and administrative expense	11,511	10,563	9%	10,469	1%
% of revenues	31.6%	30.8%		32.3%	
Interest expense (income), net	54	59	—	19	—
Other expense (income), net	66	(196)	—	(140)	—
Income before income taxes	4,325	4,886	-11%	4,623	6%
Income tax expense	2,392	646	270%	863	-25%
Effective tax rate	55.3%	13.2%		18.7%	
NET INCOME	\$ 1,933	\$ 4,240	-54%	\$ 3,760	13%
Diluted earnings per common share	\$ 1.17	\$ 2.51	-53%	\$ 2.16	16%

III. Business Strategy

STRATEGIC POSITIONING OF A BUSINESS

Understanding a company's positioning strategy is the key to understanding its business

- **Cost Leadership** – a firm sets out to become the low cost producer in its industry (example: Ikea)
- **Differentiation** – a firm seeks to create unique products that satisfy buyer needs, hence charging them a premium (example: Louis Vuitton)
- **Cost Focus** – cost leadership in a targeted segment (the firm tries to address just one part of the market) (example: Zara)
- **Differentiation focus**– differentiation in a targeted segment (the firm tries to address just one part of the market) (example: Armani Jeans)

Competitive Scope	Broad	Cost Leadership	Differentiation
	Narrow	Cost Focus	Differentiation Focus
		Lower Cost	Differentiation
Competitive Advantage			

STRATEGIC POSITIONING OF A BUSINESS

A company's financial goals should always be coherent with its strategic positioning

- **Cost Leadership** – a company is able to sell high volumes (to many people) at a low price
- **Differentiation** – a firm has a limited number of clients and sells lower volumes at a premium price
- **Cost Focus** – a company sells the cheapest product in a particular niche of the market
- **Differentiation focus**– a firm that sells a premium product to a niche client group

Cost Leadership



Volume



Price

Differentiation



Volume



Price

Cost Focus



Volume



Price

Differentiation Focus



Volume



Price

IV. Growing a Business

WHAT FACTORS INFLUENCE THE GROWTH OF SALES?

Several factors need to be considered when analyzing a company's sales

- **Macroeconomic:** The overall state of the economy in the region where the company operates
- **Industry:** Almost all industries show cyclical patterns
- **Competition:** The competitive environment that a firm faces
- **Company-specific factors:** Internal factors play an important role in a firm's future



Macroeconomic



Industry Life Cycle



Competition



Company-specific
factors

WHAT FACTORS INFLUENCE THE GROWTH OF SALES?

Macroeconomic factors

Every business operates in a complex environment, which is, in large part driven by the current Macroeconomic conditions. There aren't many businesses, if any, which can be considered independent from the Macroeconomic cycles of the economy.

For example, it will be very questionable if during a recession or an economic slowdown, a consumer driven business expects to increase its growth. Customers spend less money in such periods, and companies tend to grow slower.

Other important factors are geopolitical and legislative developments. For example, when many EU countries banned certain Russian exports, businesses in Russia were impacted very negatively.

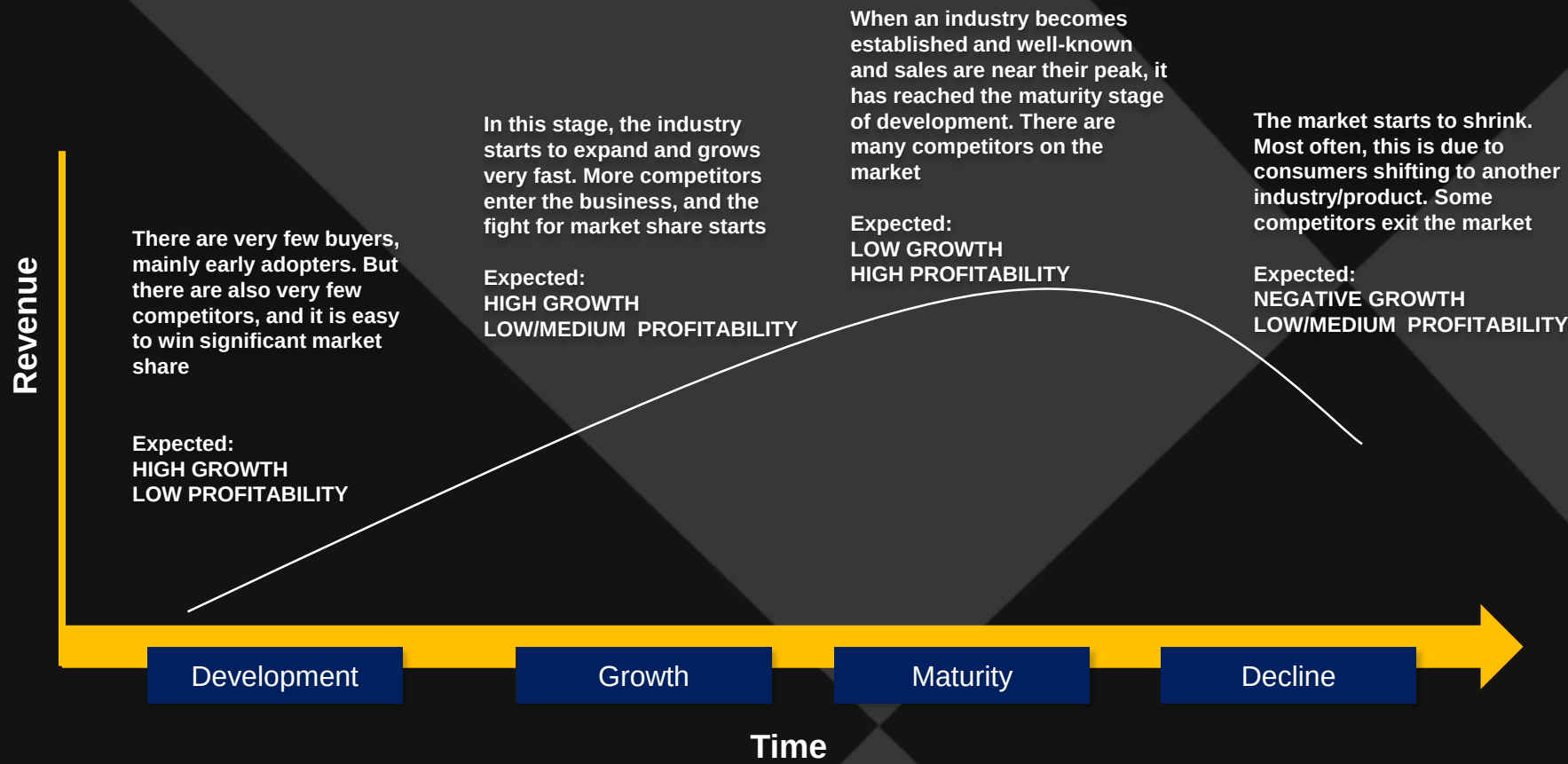


**Macroeconomic
factors**

WHAT FACTORS INFLUENCE THE GROWTH OF SALES?



The Industry Life Cycle Model



WHAT FACTORS INFLUENCE THE GROWTH OF SALES?

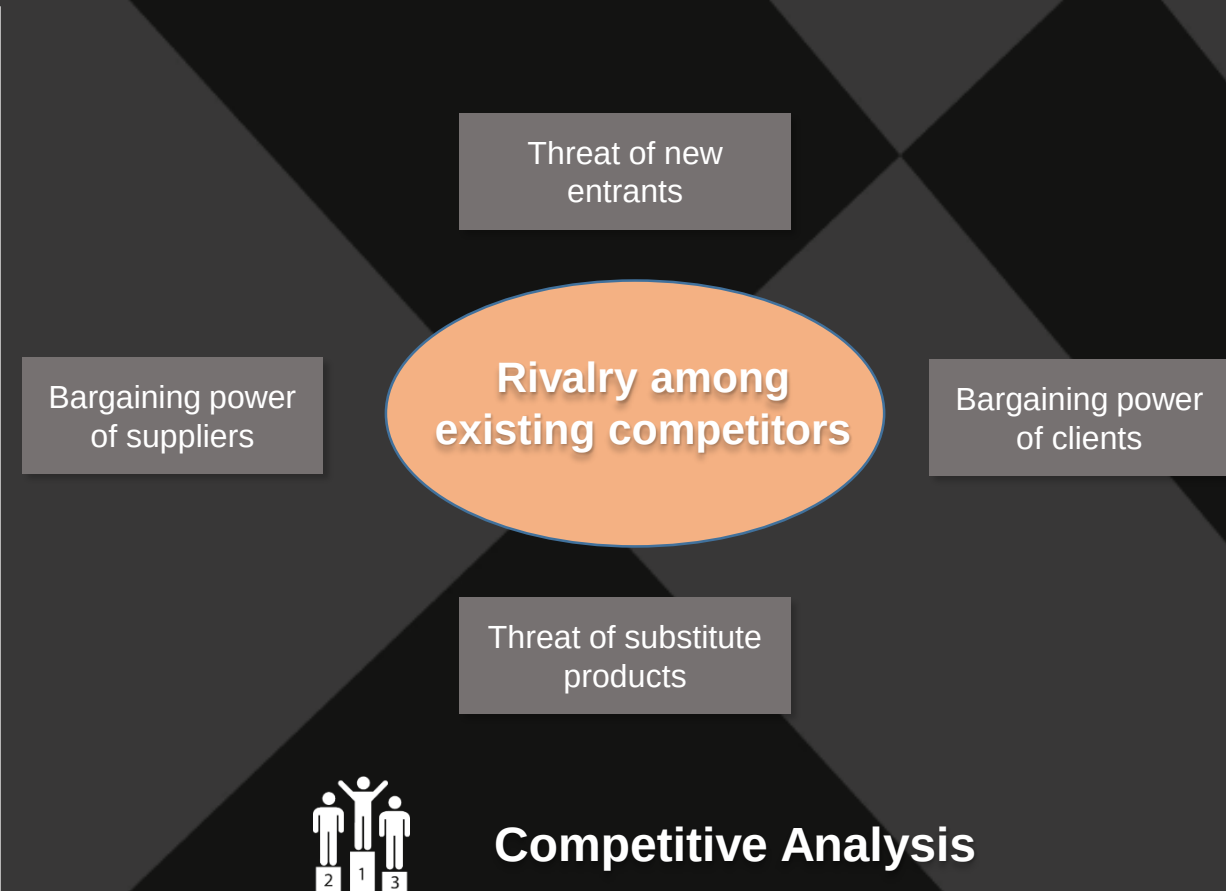
Competition

The Model that you see on the right is Michael Porter's notorious Five Forces model

It is one of the main pillars of modern competitive strategy studies

According to Porter's model, every industry is shaped by Five Forces – the threat of new entrants, the threat of substitute products, the bargaining power of suppliers, the bargaining power of clients, and the rivalry among existing competitors

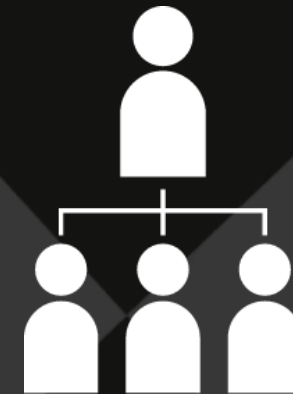
It's a strategic tool designed to give a global overview, rather than a detailed business analysis technique. It helps review the strengths of a market position, based on five key forces



WHAT FACTORS INFLUENCE THE GROWTH OF SALES?

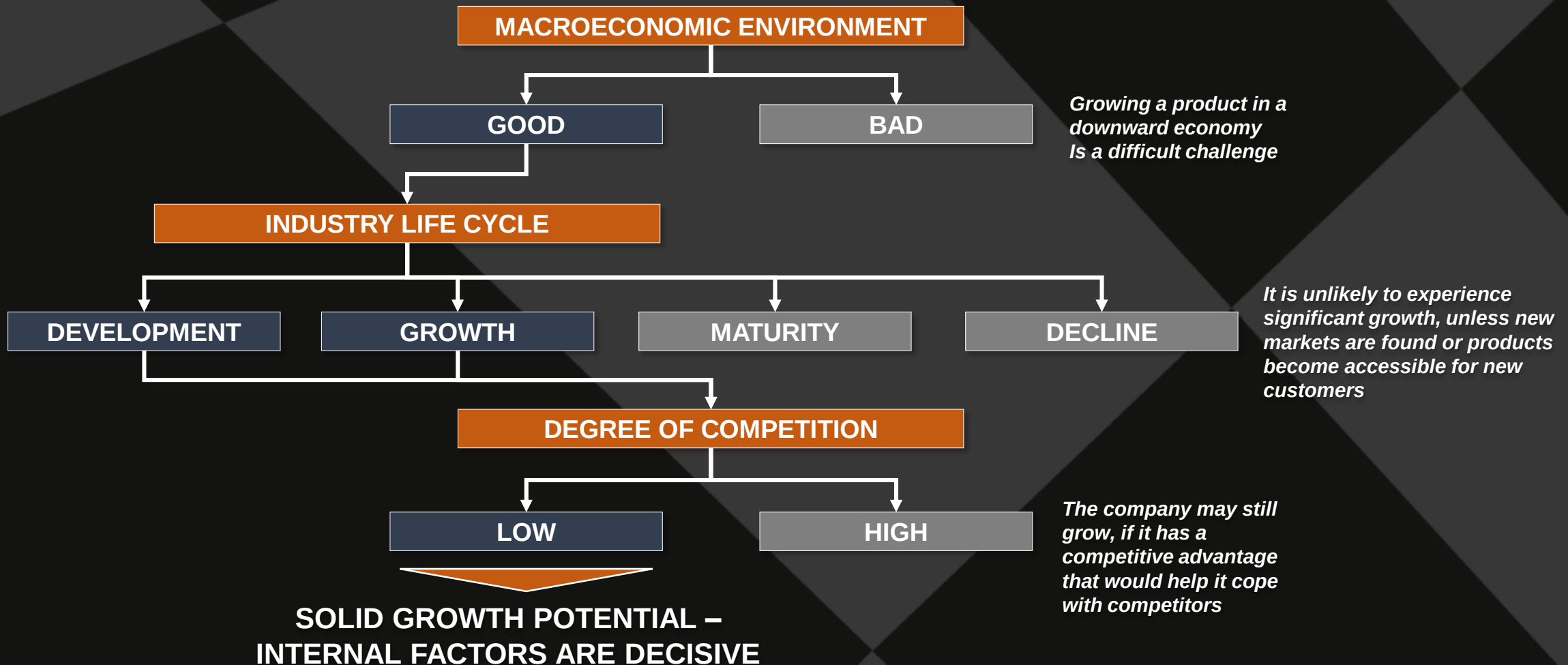
A number of internal factors can impact a company's ability to grow its sales:

- **Know-how** – what employees are able to do and know how to do; every company is only as powerful as its employees
- **Organizational culture** – some organizations are hungry and determined to succeed others are not
- **Access to resources** – financial resources, infrastructure, relationships, etc.
- **Strategic focus** – some companies know what they want to achieve and how to do it (e.g. who their customers are and how to satisfy them)
- **Brand recognition** – think of companies, like Coca Cola and P&G; a big part of their business is the fact that consumers recognize their brands



Company-specific factors

ANALYZING A COMPANY'S GROWTH POTENTIAL



V. Profitability

WHAT FACTORS INFLUENCE PROFITABILITY?

How difficult is it for a company to be profitable?

It is very difficult to build a profitable company. As you can see on the right, there are a number of costs that need to be sustained and just one stable source of income – the sale of products or services

A company has to take into account all possible expenses and then charge its clients a price that will make its business profitable

But what if its competitors are charging a lower price? In most markets, companies are price takers, which means they have to price their product at the current market price

REVENUES – COSTS = PROFITABILITY

REVENUES

Sale of products or services
Other income

COSTS

Cost of goods sold
Cost of personnel
Bonuses
Cost of services
Utility expenses
Rental expenses
Leasing expenses
D&A
Interest expenses
Extraordinary expenses
Taxes



WHAT FACTORS INFLUENCE PROFITABILITY?

Several factors need to be considered when analyzing a company's profitability

- **Supply Chain Bargaining Power:** Who are the firm's suppliers, and how likely is it that they increase the prices of their products?
- **Clients Bargaining Power:** Are clients able to negotiate lower prices?
- **Competitive Pressure:** Are there competitors, who apply downward price pressure?
- **In-house Efficiency:** Is the company able to carry out its operations in an efficient and profitable way? Are there costs that can be eliminated or reduced without compromising the quality of the company's products and the satisfaction of its employees?



**Supply Chain
Bargaining Power**



**Clients Bargaining
Power**



**Competitive
Pressure**



**In-house
Efficiency**

WHAT FACTORS INFLUENCE PROFITABILITY?

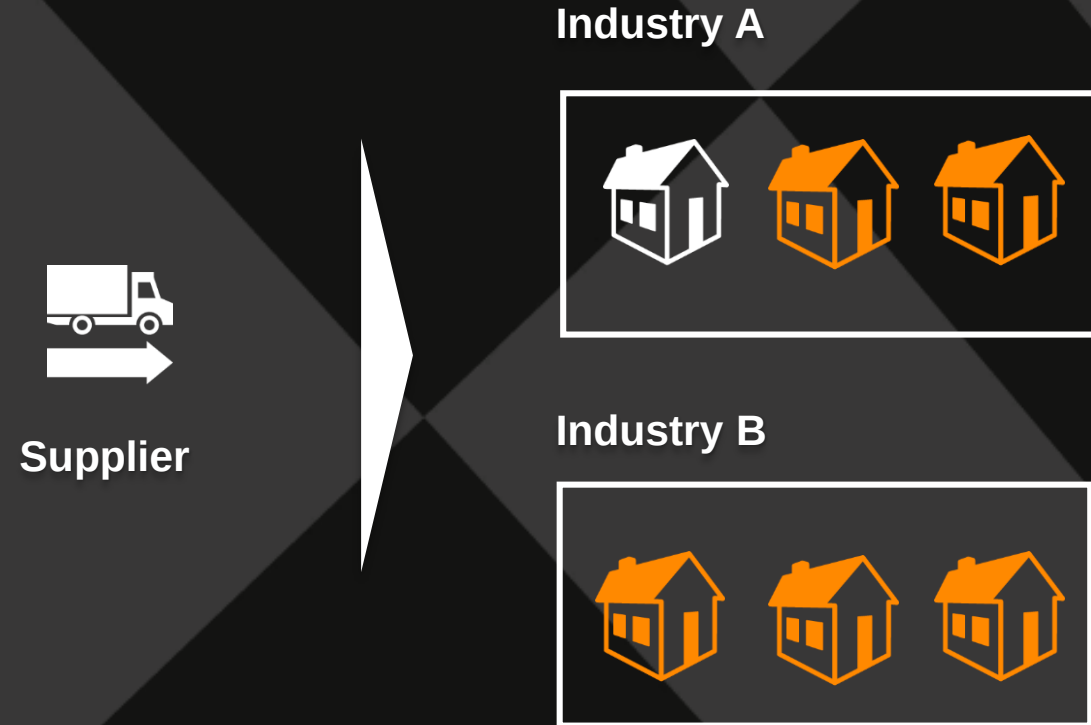
Supply Chain Bargaining Power

Most suppliers work with clients from different industries. A supplier will have strong bargaining power when he is not in danger, even if he loses a company's business

For example, Coca Cola is a supplier for many fast food chains, like McDonald's, Burger King, and Subway. But it also works with a number of different types of clients (supermarkets, hypermarkets, restaurants, night clubs, caterers, etc.)

And given that Coca Cola is able to sell its products to a large number of companies in different industries, and customers enjoy its products, it has a strong bargaining power over most (if not all) clients

Powerful suppliers are much more likely to increase the prices of their products



WHAT FACTORS INFLUENCE PROFITABILITY?

Client Bargaining Power

Clients can have bargaining power for two reasons: size and threat of using an alternative product

Large clients can put downward pressure on prices, because they know their business is important and that no company would like to lose it

The threat of using an alternative product is credible when such a product is available on the marketplace and when it satisfies the same (or similar) needs as the ones satisfied by the company's product

For example, in the energy drinks market, Red Bull and Monster are two direct alternatives, but consumers would also consider a third option – drinking coffee (a substitute product that satisfies similar needs)



WHAT FACTORS INFLUENCE PROFITABILITY?

Competitive Pressure

Competition is something with which most firms have to deal, especially in a mature industry where companies are fighting for clients and run out of ways to increase their market share

In such scenarios, some firms start to decrease the price of their products, which at the end of the day, translates into all firms reducing the prices of their products

It is proven that nobody wins from price wars, but from time to time, this occurs in some industries. This is detrimental to the profitability of all companies in the industry

Healthy competition is a good thing, but price wars are not. There is a fine line between the two



Competitive Pressure

WHAT FACTORS INFLUENCE PROFITABILITY?

In-house Efficiency

Cost of Admin Personnel, Utility Expenses, Accounting Department, Legal Department, Marketing Department, Top Management salaries, Company Cars, Rental costs, IT maintenance, Service contracts, Payroll, Taxes...

All of these are expenses that a company sustains. Whenever it manages to carry out one of these activities in a more efficient way that will save it money, a company improves its profitability

For example, many US and European companies, today outsource their IT and Administration operations in India. People there have the necessary qualification and are much cheaper to hire. This allows companies to obtain efficiency and improve their profitability



In-house Efficiency

VI. Analyzing a Company's Performance

HOW TO ANALYZE A COMPANY'S PAST PERFORMANCE

Horizontal Analysis

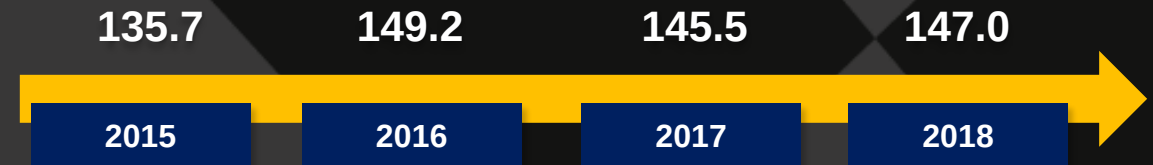
This is a type of analysis where the analyst calculates the development of a given parameter over time

For example, you should always look at the development of revenues in the last 3 or 5 years. Has there been a growth or a decreasing trend? Are there particular product categories that are growing much faster than the rest? Which products are in decline and how likely is it that the decline will continue?

There are two main formulas you need to remember in order to perform horizontal analysis: year-on-year growth and compound annual growth rate (CAGR)

Year-on-year growth shows the growth between two years, while CAGR allows us to see the average growth that was observed throughout a larger number of periods

GM's Revenue: 2009-2013



$$\text{Year-on-Year Growth \%} = \frac{\text{Revenue 2018}}{\text{Revenue 2017}} - 1 = 1.01\%$$

$$\begin{aligned} \text{CAGR} &= \left(\frac{\text{Ending Value}}{\text{Beginning Value}} \right)^{\left(\frac{1}{\text{\# of years}} \right)} - 1 \\ &= \left(\frac{147.0}{135.7} \right)^{\left(\frac{1}{4} \right)} - 1 = 2.01\% \end{aligned}$$

HOW TO ANALYZE A COMPANY'S PAST PERFORMANCE

Vertical Analysis

The term, Vertical Analysis, is the proportional analysis of a financial statement, where each line item is listed as a percentage of another item.

As you can see in the example on the right, below Gross Profit, EBIT, and Net Income have been calculated to show the portion of Revenues they represent. These percentages allow us to have an idea about the weight of each of these subtotals with respect to the total.

This P&L allows us to gain an idea about GM's margins and profitability. In 2013, the firm makes a Gross Profit of 13.2% and its Net Income Margin is equal to 3.4%. This is a troubled business that has had significant difficulties in recent years.

Vertical analysis helps us understand how a firm makes its money and how efficient it is in carrying out its business operations

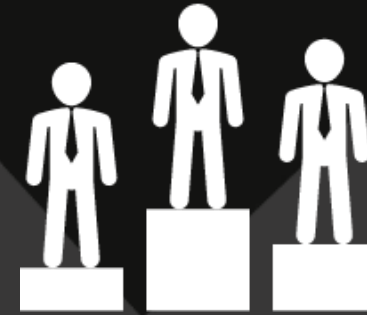
Key P&L Items (\$bn)	2009A	2010A	2011A	2012A	2013A
Automotive	104.6	135.3	148.9	150.3	152.1
GM Financial	-	0.3	1.4	2.0	3.3
Total Revenues	104.6	135.6	150.3	152.3	155.4
Cost of Sales	(112.2)	(118.8)	(130.4)	(140.2)	(134.9)
Gross Profit	(7.6)	16.8	19.9	12.0	20.5
as a % of Revenues	-7.3%	12.4%	13.2%	7.9%	13.2%
Operating expenses	(13.4)	(11.7)	(12.9)	(15.2)	(14.8)
EBIT	(21.0)	5.1	6.9	(3.2)	5.7
as a % of Revenues	-20.1%	3.7%	4.6%	-2.1%	3.6%
Interest income/(expense)	(6.1)	(1.1)	(0.5)	(0.5)	(0.3)
Extraordinary items	130.2	3.2	2.8	(25.0)	2.1
Minorities	(0.4)	(0.3)	(0.1)	0.1	0.0
Taxes	2.2	(0.7)	0.1	34.8	(2.1)
Net Income	104.8	6.2	9.2	6.2	5.3
as a % of Revenues	100.2%	4.6%	6.1%	4.1%	3.4%

HOW TO ANALYZE A COMPANY'S PAST PERFORMANCE

Peer comparison

The most meaningful analysis you can carry out is the one that compares the firm you are analyzing and three, four, or five of its peers

Perform horizontal and vertical analysis for the financials of each of these firms and then calculate an average. This would allow you to understand whether the firm you are analyzing performs above or below the average of its competitor firms. In case it outperforms them, you will need to understand why. Perhaps, the firm has a competitive advantage, or there is lack of competition in the industry. The same thing goes for the scenario when the company underperforms its peers; you will have to understand the reason for that. Operational inefficiencies? Slower growth? Unprofitable products?



Peer Comparison

HOW TO ANALYZE A COMPANY'S PAST PERFORMANCE

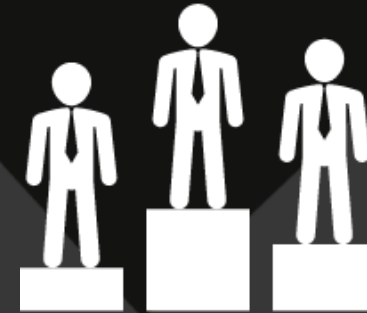
Selecting comparable firms

Peer analysis sounds like a great idea, but we have to determine which companies are comparable to the one we are analyzing

Make a list of the companies in the industry and compare the following:

- Size (in terms of revenues, employees, assets)
- Product portfolio
- Geographical reach
- Strategic Positioning (cost leadership vs differentiation)
- Maturity
- Type of clients served
- Distribution channels

Select three or four companies that are a close match to the target company and use them in order to calculate an average



Peer Comparison

ANALYZING THE MAIN LINE ITEMS

Revenues

This is the top line of a company's Profit & Loss statement, and it shows the amount of money that the firm received for selling products or services

We can analyze sales in a number of ways:

- **Geographical Footprint** – will show us fast/slow growth regions, which allows taking responsive action or creating expectations about future development
- **Product Breakdown** – see which are the firm's top selling products and whether their sales are accelerating or slowing down
- **Distribution Channels** – different distribution channels have different marginality and sales volumes
- **Key Clients** – how big are the firm's largest clients and for what portion of revenues do they account for



**Geographical
Footprint**



**Product
Breakdown**



**Distribution
Channels**



**Key
Clients**

ANALYZING THE MAIN LINE ITEMS

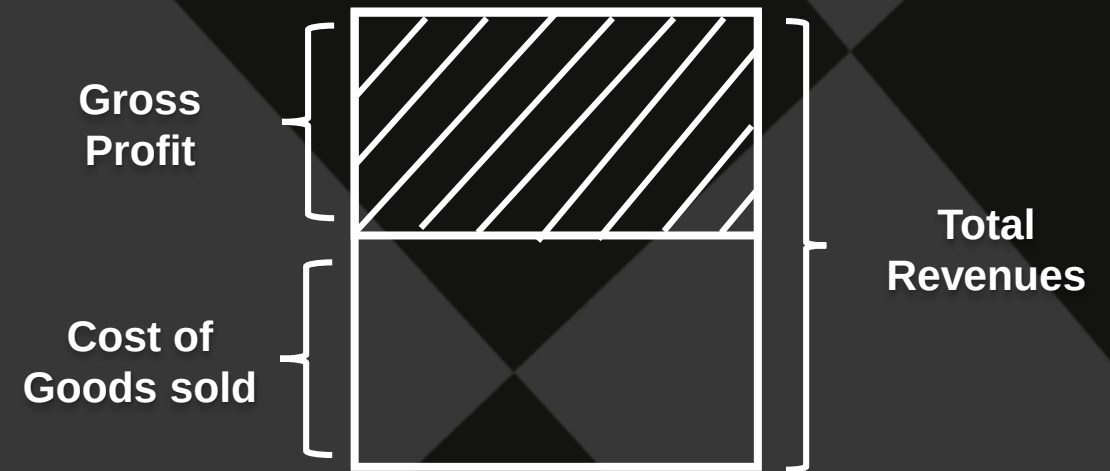
Gross Profit

Gross Profit is what remains when we subtract the cost of the goods that were sold from revenues. It is a measure that gives us an idea of how much the company earns from its operations before covering operating expenses

A negative Gross Profit is very alarming

We can analyze Gross Profit in terms of:

- Profitability per product – shows which products are profitable and the floor price that can be provided to clients
- Profitability per client – how much does the firm make from each of its clients; can serve as a leverage in negotiations
- Profitability per segment – allows us to see how various segments of the firm contribute to the total figures



ANALYZING THE MAIN LINE ITEMS

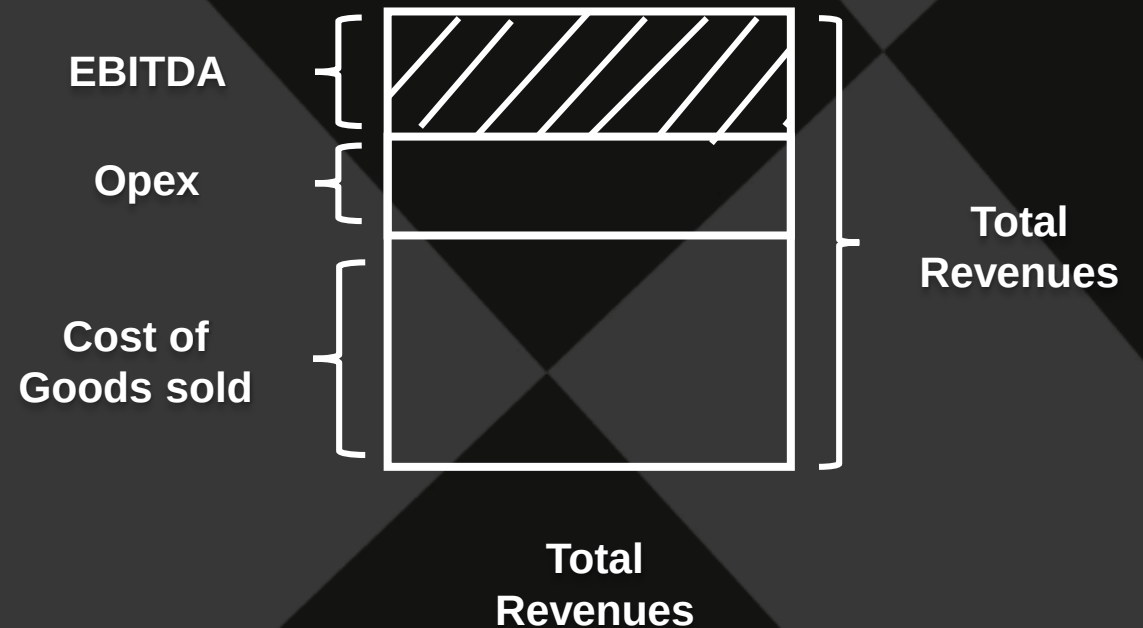
EBITDA

The abbreviation stands for Earnings before Interest, Taxes, Depreciation and Amortization. EBITDA is one of the most important subtotals in a company's P&L. It shows the company's profitability after taking into consideration Cogs and Operating expenses

EBITDA is believed to show the true development of the operating business, as it does not take into consideration costs that do not occur in the normal course of business

A negative EBITDA shows that a company has big problems with its core business and operating cash flows

There are many analysts who prefer to use EBIT, instead of EBITDA, as a measurement for operating profits. It takes into consideration D&A, which is a non-cash expense



ANALYZING THE MAIN LINE ITEMS

Net Income

We have arrived to the bottom line of the Profit & Loss statement – Net Income

This is the profit or loss the company has generated after all expenses are taken into consideration

And, of course, it deserves some of our attention, as this is the amount that will either be distributed to shareholders (as dividends) or reinvested in the business

An analyst could be interested in learning:

- What portion of EBIT is converted into Net Income (is the firm spending a lot for interests, taxes, one-off events, etc.)
- If there have been one-off events are they polluting the overall quality of earnings
- What is the return on equity (ROE) that the company's shareholders receive compared with an industry average?

Profit & Loss

Revenues

- Cost of goods sold
- = Gross profit
- Operating expenses
- = EBITDA
- D&A
- = EBIT
- Interest expense
- Extraordinary items
- = EBT
- Taxes
- = Net Income

ANALYZING THE MAIN LINE ITEMS

Financial Leverage & Interest Expenses

A firm needs to be able to keep a fine balance between the amount of debt that it borrows from lenders and its own funds. If it does not have a sufficient amount of own funds with respect to its debts, its interest rate could become higher, which in turn will be a problem for its profitability and cash flow

A good analyst should look at:

- Company's borrowing rate ($\text{Interest expenses} / \text{Financial Debt}$)
- Leverage ($\text{Financial Debt} / \text{Equity}$)
- Leverage of comparable companies
- Coverage ratio ($\text{EBITDA} / \text{Interest expenses}$)

Financial
Debt

vs.

Own
Funds
(Equity)



ANALYZING THE MAIN LINE ITEMS

Working Capital

Working Capital is the amount of money that stays “locked” in a company’s business. The firm needs to use the money in order to carry out its business operations. A high level of Working Capital is a significant investment. That is why a lot of analysts are interested in this component of a firm’s business. As you can see on the right, working capital is formed by three components: accounts receivable, inventory, and payables

We have to analyze several aspects of a company’s Working Capital:

- How much is invested?
- Is there a seasonality pattern?
- Is there an increasing trend?

$$\text{Working Capital} = \text{Receivables} + \text{Inventory} - \text{Payables}$$

Trade Receivables = Money owed by customers

Trade Payable = Money owed to suppliers

Inventory = The value of goods in the warehouse

$$\text{DSO} = \frac{\text{Accounts Receivable}}{\text{Revenues}} \times 365$$

$$\text{DPO} = \frac{\text{Trade Payables}}{\text{Cogs}} \times 365$$

$$\text{DIO} = \frac{\text{Inventory}}{\text{Cogs}} \times 365$$

ANALYZING THE MAIN LINE ITEMS

Operating and Net Cash Flow

Generating Cash Flow is very important. A business can be profitable, but still run out of cash and thus go bankrupt

All analysts are interested in the amount of cash that a firm makes

There is an important difference between Operating and Net Cash Flow. Operating Cash Flow takes into consideration all cash inflows and outflows that are part of a firm's core operations (and are related to its main business)

At the same time Net Cash Flow includes all cash movements – operating and non-operating

Analysts typically pay more attention to a firm's operating cash flows and try to understand if there are any other cash outflows that can be discontinued through a tax, cost, or financial optimization



Operating and Net Cash Flow

SWOT ANALYSIS

SWOT analysis helps you evaluate a company's future prospects

As you can see on the right, this is a matrix that is divided into four parts. It describes a company's strengths, weaknesses, opportunities, and threats. The analyst looks inside the company's business and defines its strengths and weaknesses. After that, he puts the company's business into perspective and evaluates potential opportunities and threats (industry trends, potential new markets, new applications of existing products, new entry of competitor firms, substitute products, etc.).

SWOT analysis is a favorite tool for most high-level executives because it provides clarity and it is to the point

Internal	Strengths	Weaknesses
	Opportunities	Threats
External	Helpful	Harmful